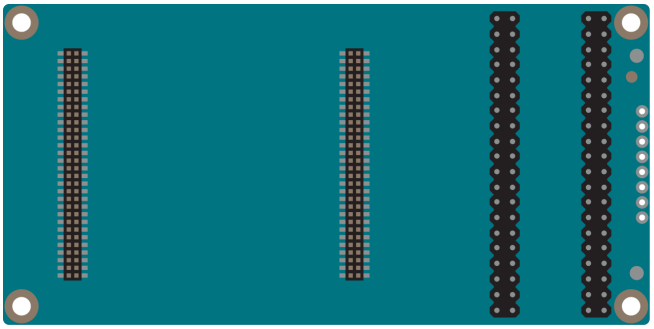
User Manual
SKU: ASX00085



Description

The Arduino UNO Breakout Carrier is designed to give developers complete, direct access to every signal available on the UNO Q's JMEDIA and JMISC high-speed connectors. Ideal for advanced prototyping, testing, and integration work, it exposes all lines — including high-speed video, camera, audio, I²C, SPI, UART, PWM, power rails, and control signals — to clearly labeled, easy-to-use breakout headers.

Features

- Connectors
  - Two 2x20 Male Headers (2.54 mm)
  - Two 2x30 Male Headers(JMEDIA and JMISC 1.27 mm)
  - 1x8 through-hole pads (2.54 mm)
- Power
  - Powered from the host UNO Q
  - VIN input power rails (+7-24 VDC)
- I/O
  - I2C
  - Microphone In / Headphone Out
  - Earphone Out
  - Audio Line Out
  - PWM
  - PSSI
  - GPIO

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1 The Board

1.1 Related Products

- Arduino UNO Q (SKU: ABX00162 - ABX00173)

2 Ratings

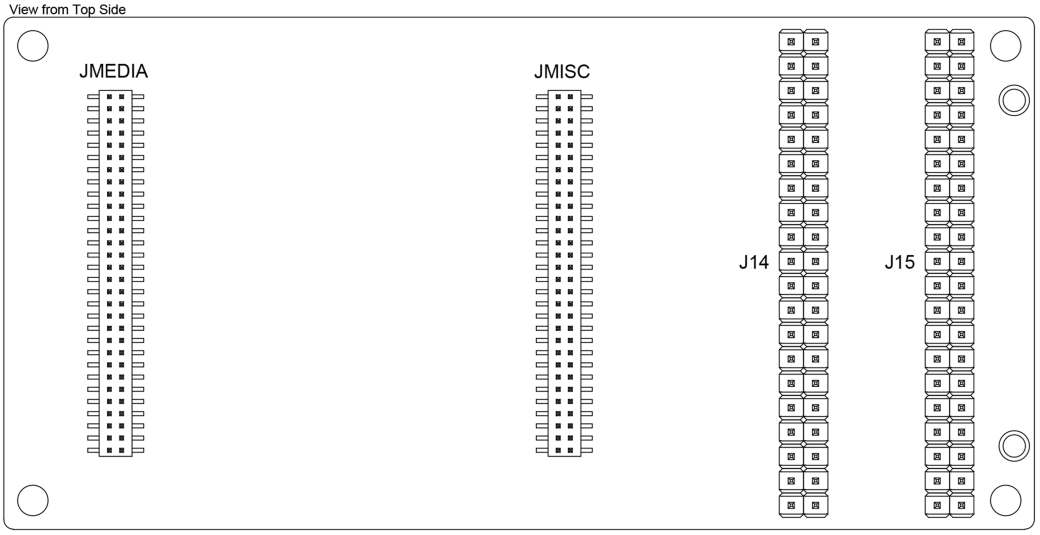
2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
T	Conservative thermal limits	-10	20	60	°C
V_{IN}	Input voltage from VIN pad	6	7	24	V

3 Functional Overview

The UNO Breakout Carrier expands the connectivity of the Arduino UNO Q, featuring a variety of 2.54 mm male header connectors.

3.1 Board Topology

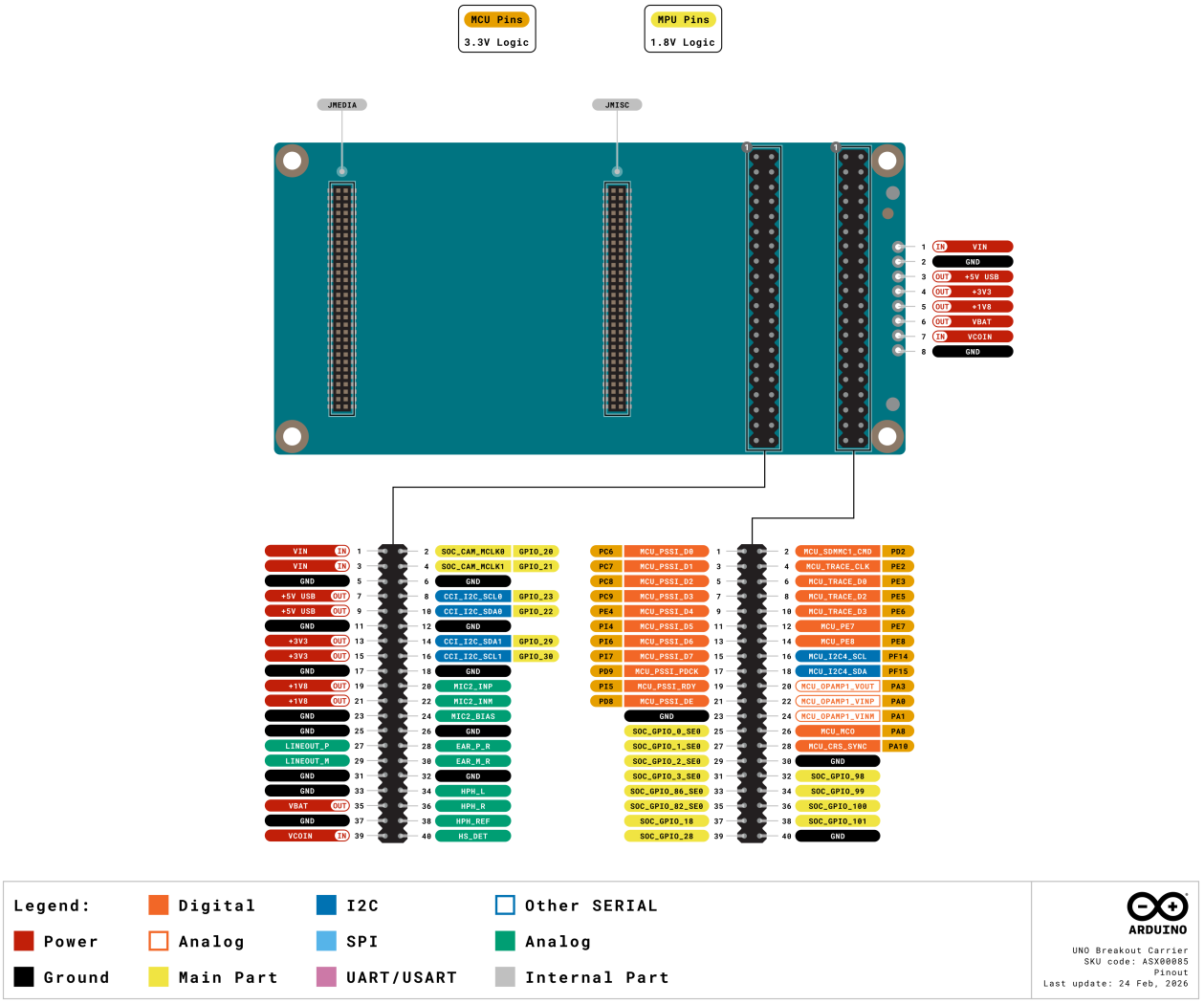


Top view - connectors

Ref.	Description
J14	Male header connector 2x20 2.54 mm
J15	Male header connector 2x20 2.54 mm
JMEDIA	High-speed male header connector 2x30 1.27 mm
JMISC	High-speed male header connector 2x30 1.27 mm

3.2 Pinout

The UNO Breakout Carrier pinout is shown in the following figure.



UNO Breakout Pinout

3.2.1 J14

Pin	Function	Type	Description
1	VIN	Power In	Voltage Input
2	GPIO_20	Digital	SOC_CAM_MCLK0
3	VIN	Power In	Voltage Input
4	GPIO_21	Digital	SOC_CAM_MCLK1
5	GND	Ground	Ground
6	GND	Ground	Ground
7	+5V USB	Power Out	+5V USB Power Output
8	GPIO_23	Digital / I2C	CCI_I2C_SCL0
9	+5V USB	Power Out	+5V USB Power Output
10	GPIO_22	Digital / I2C	CCI_I2C_SDA0
11	GND	Ground	Ground
12	GND	Ground	Ground
13	+3V3	Power Out	+3.3V Power Output
14	GPIO_29	Digital / I2C	CCI_I2C_SDA1
15	+3V3	Power Out	+3.3V Power Output
16	GPIO_30	Digital / I2C	CCI_I2C_SCL1
17	GND	Ground	Ground
18	GND	Ground	Ground
19	+1V8	Power Out	+1.8V Power Output
20	MIC2_INP	Analog	Microphone Input Positive
21	+1V8	Power Out	+1.8V Power Output
22	MIC2_INM	Analog	Microphone Input Negative
23	GND	Ground	Ground
24	MIC2_BIAS	Analog	Microphone Bias
25	GND	Ground	Ground
26	GND	Ground	Ground
27	LINEOUT_P	Analog	Audio Line Out Positive
28	EAR_P_R	Analog	Ear Right Positive
29	LINEOUT_M	Analog	Audio Line Out Negative
30	EAR_M_R	Analog	Ear Right Negative
31	GND	Ground	Ground
32	GND	Ground	Ground
33	GND	Ground	Ground
34	HPH_L	Analog	Headphone Left
35	VBAT	Power Out	+3.8V Buck Converter Output
36	HPH_R	Analog	Headphone Right
37	GND	Ground	Ground
38	HPH_REF	Analog	Headphone Reference
39	VCOIN	Power In	Coin Cell / RTC Backup Voltage Input
40	HS_DET	Analog	Headphone Detection

3.2.2 J15

Pin	Function	Type	Description
1	MCU_PSSI_D0 / PC6	Digital	MCU GPIO
2	MCU_SDMMC1_CMD / PD2	Digital	MCU GPIO
3	MCU_PSSI_D1 / PC7	Digital	MCU GPIO
4	MCU_TRACE_CLK / PE2	Digital	MCU GPIO
5	MCU_PSSI_D2 / PC8	Digital	MCU GPIO
6	MCU_TRACE_D0 / PE3	Digital	MCU GPIO
7	MCU_PSSI_D3 / PC9	Digital	MCU GPIO
8	MCU_TRACE_D2 / PE5	Digital	MCU GPIO
9	MCU_PSSI_D4 / PE4	Digital	MCU GPIO
10	MCU_TRACE_D3 / PE6	Digital	MCU GPIO
11	MCU_PSSI_D5 / PI4	Digital	MCU GPIO
12	MCU_PE7 / PE7	Digital	MCU GPIO
13	MCU_PSSI_D6 / PI6	Digital	MCU GPIO
14	MCU_PE8 / PE8	Digital	MCU GPIO
15	MCU_PSSI_D7 / PI7	Digital	MCU GPIO
16	MCU_I2C4_SCL / PF14	Digital / I2C	MCU GPIO
17	MCU_PSSI_PDCK / PD9	Digital	MCU GPIO
18	MCU_I2C4_SDA / PF15	Digital / I2C	MCU GPIO
19	MCU_PSSI_RDY / PI5	Digital	MCU GPIO
20	MCU_OPAMP1_VOUT / PA3	Analog	MCU GPIO / OPAMP OUT
21	MCU_PSSI_DE / PD8	Digital	MCU GPIO
22	MCU_OPAMP1_VINP / PA0	Analog	MCU GPIO / OPAMP IN +
23	GND	Ground	Ground
24	MCU_OPAMP1_VINM / PA1	Analog	MCU GPIO / OPAMP IN -
25	SOC_GPIO_0_SE0	Digital	MPU GPIO
26	MCU_MCO / PA8	Digital	MCU GPIO
27	SOC_GPIO_1_SE0	Digital	MPU GPIO
28	MCU_CRS_SYNC / PA10	Digital	MCU GPIO
29	SOC_GPIO_2_SE0	Digital	MPU GPIO
30	GND	Ground	Ground
31	SOC_GPIO_3_SE0	Digital	MPU GPIO
32	SOC_GPIO_98	Digital	MPU GPIO
33	SOC_GPIO_86_SE0	Digital	MPU GPIO
34	SOC_GPIO_99	Digital	MPU GPIO
35	SOC_GPIO_82_SE0	Digital	MPU GPIO
36	SOC_GPIO_100	Digital	MPU GPIO
37	SOC_GPIO_18	Digital	MPU GPIO
38	SOC_GPIO_101	Digital	MPU GPIO
39	SOC_GPIO_20	Digital	MPU GPIO
40	GND	Ground	Ground

3.2.3 JMEDIA

Pin	Function	Type	Description
1	GND	Ground	Ground
2	GND	Ground	Ground
3	NC	None	Not Connected
4	NC	None	Not Connected
5	NC	None	Not Connected
6	NC	None	Not Connected
7	GND	Ground	Ground
8	GND	Ground	Ground
9	NC	None	Not Connected
10	NC	None	Not Connected
11	GND	Ground	Ground
12	NC	None	Not Connected
13	GND	Ground	Ground
14	GND	Ground	Ground
15	NC	None	Not Connected
16	SOC_CAM_MCLK0 / GPIO_20	Digital	MPU GPIO
17	NC	None	Not Connected
18	SOC_CAM_MCLK1 / GPIO_21	Digital	MPU GPIO
19	GND	Ground	Ground
20	GND	Ground	Ground
21	NC	None	Not Connected
22	CCI_I2C_SDA1 / GPIO_29	I2C	MPU GPIO
23	NC	None	Not Connected
24	CCI_I2C_SCL1 / GPIO_30	I2C	MPU GPIO
25	GND	Ground	Ground
26	GND	Ground	Ground
27	NC	None	Not Connected
28	NC	None	Not Connected
29	NC	None	Not Connected
30	NC	None	Not Connected
31	GND	Ground	Ground
32	GND	Ground	Ground
33	NC	None	Not Connected
34	NC	None	Not Connected
35	NC	None	Not Connected
36	NC	None	Not Connected
37	GND	Ground	Ground
38	GND	Ground	Ground
39	NC	None	Not Connected
40	NC	None	Not Connected
41	NC	None	Not Connected

Pin	Function	Type	Description
42	NC	None	Not Connected
43	GND	Ground	Ground
44	GND	Ground	Ground
45	NC	None	Not Connected
46	NC	None	Not Connected
47	NC	None	Not Connected
48	NC	None	Not Connected
49	GND	Ground	Ground
50	GND	Ground	Ground
51	NC	None	Not Connected
52	CCI_I2C_SCL0 / GPIO_23	I2C	MPU GPIO
53	NC	None	Not Connected
54	CCI_I2C_SDA0 / GPIO_22	I2C	MPU GPIO
55	GND	Ground	Ground
56	GND	Ground	Ground
57	VIN	Power In	Voltage Input
58	+3V3	Power Out	+3.3V Power Output
59	VIN	Power In	Voltage Input
60	+3V3	Power Out	+3.3V Power Output

3.2.4 JMISC

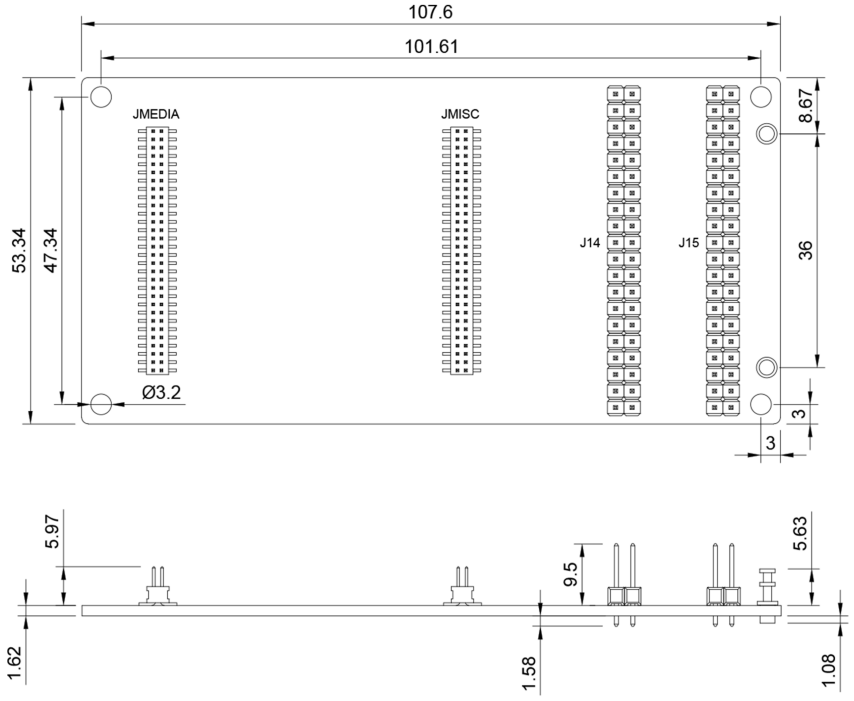
Pin	Function	Type	Description
1	MCU_PSSI_D0 / PC6	Digital	MCU GPIO
2	MCU_SDMMC1_CMD / PD2	Digital	MCU GPIO
3	MCU_PSSI_D1 / PC7	Digital	MCU GPIO
4	MCU_TRACE_CLK / PE2	Digital	MCU GPIO
5	MCU_PSSI_D2 / PC8	Digital	MCU GPIO
6	MCU_TRACE_D0 / PE3	Digital	MCU GPIO
7	MCU_PSSI_D3 / PC9	Digital	MCU GPIO
8	MCU_TRACE_D2 / PE5	Digital	MCU GPIO
9	MCU_PSSI_D4 / PE4	Digital	MCU GPIO
10	MCU_TRACE_D3 / PE6	Digital	MCU GPIO
11	MCU_PSSI_D5 / PI4	Digital	MCU GPIO
12	MCU_PE7 / PE7	Digital	MCU GPIO
13	MCU_PSSI_D6 / PI6	Digital	MCU GPIO
14	MCU_PE8 / PE8	Digital	MCU GPIO
15	MCU_PSSI_D7 / PI7	Digital	MCU GPIO
16	MCU_I2C4_SCL / PF14	Digital / I2C	MCU GPIO
17	MCU_PSSI_PDCK / PI9	Digital	MCU GPIO
18	MCU_I2C4_SDA / PF15	Digital / I2C	MCU GPIO
19	MCU_PSSI_RDY / PI5	Digital	MCU GPIO
20	MCU_OPAMP1_VOUT / PA3	Analog	MCU GPIO / OPAMP OUT

Pin	Function	Type	Description
21	MCU_PSSI_DE / PD8	Digital	MCU GPIO
22	MCU_OPAMP1_VINP / PA0	Analog	MCU GPIO / OPAMP IN +
23	MCU_MCO / PA8	Digital	MCU GPIO
24	MCU_OPAMP1_VINM / PA1	Analog	MCU GPIO / OPAMP IN -
25	MCU_CRS_SYNC / PA10	Digital	MCU GPIO
26	GND	Ground	Ground
27	GND	Ground	Ground
28	EAR_P_R	Analog	Ear Right Positive
29	MIC2_INP	Analog	Microphone Input Positive
30	EAR_M_R	Analog	Ear Right Negative
31	MIC2_INM	Analog	Microphone Input Negative
32	LINEOUT_P	Analog	Audio Line Out Positive
33	MIC2_BIAS	Analog	Microphone Bias
34	LINEOUT_M	Analog	Audio Line Out Negative
35	GND	Ground	Ground
36	HPH_L	Analog	Headphone Left
37	SOC_GPIO_0_SE0	Digital	MPU GPIO
38	HPH_R	Analog	Headphone Right
39	SOC_GPIO_1_SE0	Digital	MPU GPIO
40	HPH_REF	Analog	Headphone Reference
41	SOC_GPIO_2_SE0	Digital	MPU GPIO
42	HS_DET	Analog	Headset Detection
43	SOC_GPIO_3_SE0	Digital	MPU GPIO
44	GND	Ground	Ground
45	SOC_GPIO_86_SE0	Digital	MPU GPIO
46	SOC_GPIO_98	Digital	MPU GPIO
47	SOC_GPIO_82_SE0	Digital	MPU GPIO
48	SOC_GPIO_99	Digital	MPU GPIO
49	SOC_GPIO_18	Digital	MPU GPIO
50	SOC_GPIO_100	Digital	MPU GPIO
51	SOC_GPIO_28	Digital	MPU GPIO
52	SOC_GPIO_101	Digital	MPU GPIO
53	+3V3	Power Out	+3.3V Power Output
54	+5V USB	Power Out	+5V USB Power Output
55	+3V3	Power Out	+3.3V Power Output
56	+5V USB	Power Out	+5V USB Power Output
57	+1V8	Power Out	+1.8V Power Output
58	GND	Ground	Ground
59	VCOIN	Power In	Coin Cell / RTC Backup Voltage Input
60	VBAT	Power Out	+3.8V Buck Converter Output

4 Mechanical Information

4.1 Board Dimensions

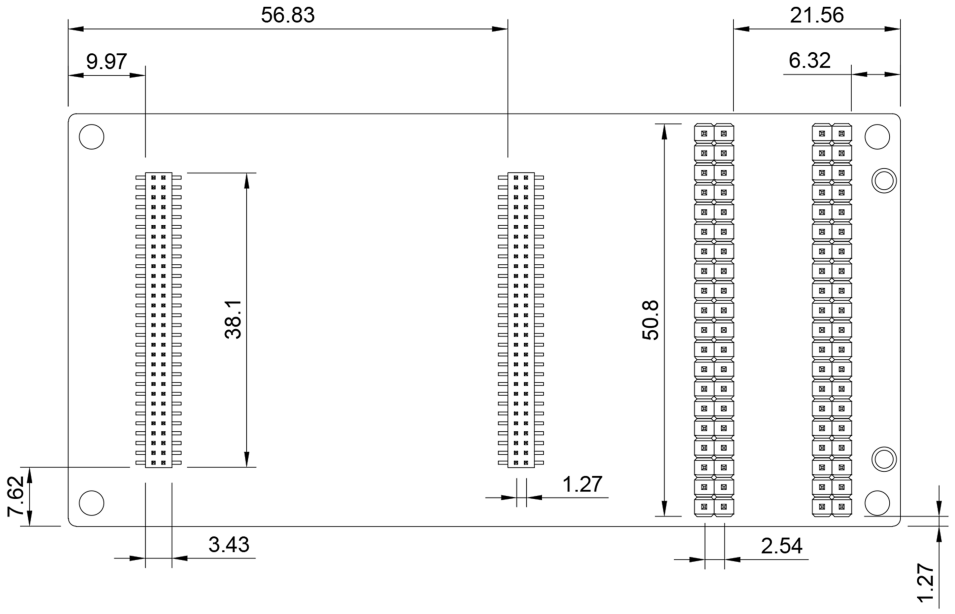
The outline and dimensions of the UNO Breakout Carrier and mounting holes can be seen in the following figure; all the dimensions are in mm.



Board outline and screw holes

4.2 Board Connectors

The UNO Breakout Carrier's connectors are placed on the top side of the board, as shown in the figure below; all the dimensions are in mm.



Mechanical View of UNO Breakout Carrier's Connectors

5 Company Information

Company name	Arduino S.r.l.
Company Address	Via Andrea Appiani, 25 - 20900 MONZA (Italy)

6 Reference Documentation

Ref	Link
Arduino IDE (Desktop)	https://www.arduino.cc/en/Main/Software
Arduino IDE (Cloud)	https://create.arduino.cc/editor
Cloud IDE Getting Started	https://create.arduino.cc/projecthub/Arduino_Genuino/getting-started-with-arduino-web-editor-4b3e4a
Arduino Pro Website	https://www.arduino.cc/pro
Project Hub	https://create.arduino.cc/projecthub?by=part&part_id=11332&sort=trending
Library Reference	https://www.arduino.cc/reference/en/
Online Store	https://store.arduino.cc/

7 Change Log

Date	Revision	Changes
06/03/2026	1	First Release